

Sparse Node Perturbation: Accuracy vs. Active-Node Count

ANP and DANP — FPGA-Oriented Motivation

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On FPGA hardware, each perturbed node requires independent noise injection and a separate weight update, making perturbation cost proportional to the number of active nodes. Restricting perturbation to a sparse subset of nodes is therefore a natural strategy for reducing the perturbation-related workload in a future hardware implementation. This report analyses the relationship between active-node count and classification accuracy under two algorithms: Activity-based Node Perturbation (ANP) and its decorrelated variant (DANP). Actual FPGA resource utilisation, latency, power consumption, and throughput have not yet been measured; this analysis characterises only the active-node fraction and the associated accuracy cost observed in software experiments. All reported values correspond to completed experimental runs stored in the repository. Unless stated otherwise, the sparse policy is *Scheduled 25%* (round-robin, 25% active nodes per layer), MNIST, 20 epochs, 3 random seeds.

1 Perturbation Operation Count Reduction

In ANP/DANP, each training step includes the following operations. The *clean forward pass* runs unconditionally and is not reduced by sparsity. The remaining operations involve only the active (perturbed) nodes k :

1. **Noise generation:** one random sample per active node per layer $\rightarrow O(k)$ per layer
2. **Activity-difference computation:** $\Delta \mathbf{a}^{(l)} = \mathbf{a}_{\text{noisy}}^{(l)} - \mathbf{a}_{\text{clean}}^{(l)}$ for active nodes $\rightarrow O(k)$ per layer
3. **Weight updates associated with perturbed nodes:** gradient contributions from active nodes $\rightarrow O(k \cdot D_l)$ per layer
4. **Optimizer updates associated with perturbed nodes:** one Adam step per updated parameter $\rightarrow O(k \cdot D_l)$ per layer

With active fraction $\alpha = k/H$, operations 1–4 involve a fraction α of the nodes relative to full perturbation. The clean forward pass is unchanged. At $\alpha = 0.25$, sparse masks imply an **approximately 75% reduction** in the number of perturbation-related operations (noise generation, activity-difference processing, and weight and optimizer updates for perturbed nodes). This cost structure is the same for ANP and DANP; only the gradient formula differs.

The active-node counts are not exact multiples of 25% because `num_active_units` applies banker’s rounding, and the 10-node output layer rounds to 2 active nodes at 25%.

Table 1: Active-node count and implied perturbation operation reduction per architecture (Scheduled 25%, MNIST). Rel. Cost = k/H_{total} ; Implied Op. Reduction = $1 - \text{Rel. Cost}$. Values are the same for ANP and DANP.

Architecture	Total Nodes H	Active Nodes k	Rel. Cost	Op. Reduction
[64, 64, 64]	202	50	0.2475	75.2%
[256, 256, 256]	778	194	0.2494	75.1%
[1024, 1024, 1024]	3082	770	0.2498	75.0%

Table 2: DANP accuracy and accuracy drop at $\approx 25\%$ active nodes (MNIST, Scheduled 25%, 20 epochs, 3 seeds).

Architecture	Full Acc.	Sparse Acc.	Acc. Drop	Rel. Cost
[64, 64, 64]	0.9391	0.9342	-0.0049	0.2475
[256, 256, 256]	0.9530	0.9480	-0.0050	0.2494
[1024, 1024, 1024]	0.9642	0.9566	-0.0076	0.2498

2 DANP Results

Across all three widths, sparse masks activate approximately 25% of nodes while accuracy drops by at most 0.76 pp. The drop is consistent across widths and does not grow substantially with network size, suggesting that the accuracy-active-node tradeoff is not limited to a specific model size.

3 ANP Results

ANP omits the decorrelation update applied in DANP. The sparse perturbation framework is identical: the same mask types, the same noise-injection mechanism, and the same update rule apply to ANP without modification.

3.1 MNIST Width Sweep

Table 3: ANP accuracy and accuracy drop at $\approx 25\%$ active nodes (MNIST, Scheduled 25%, 20 epochs, 3 seeds).

Architecture	Full ANP	Sparse ANP	Acc. Drop	Rel. Cost
[64, 64, 64]	0.9266	0.9253	-0.0013	0.2475
[256, 256, 256]	0.9223	0.9170	-0.0053	0.2494
[1024, 1024, 1024]	0.9069	0.8920	-0.0149	0.2498

Key result. At H64 and H256, Scheduled Sparse ANP preserves nearly all of the full-ANP accuracy while activating only 25% of the nodes: accuracy drops of 0.13 pp and 0.53 pp respectively are observed, both well under 0.6 pp. These results suggest that sparse ANP is viable at smaller and medium widths even without the decorrelation update.

At H1024, the drop increases to 1.49 pp, roughly double the DANP figure of 0.76 pp. The full ANP baseline is also lower than DANP across all widths, consistent with decorrelation providing a learning-efficiency advantage that becomes more pronounced at larger scale.

3.2 Random vs. Scheduled Policy at H1024

Table 4: Effect of sparse policy on ANP accuracy at H1024 (MNIST, 20 epochs, 3 seeds). Both sparse runs use 770/3082 active nodes.

Policy	Peak Accuracy	Active/Total Nodes
Full ANP	0.9069	3082/3082
Random Sparse 25%	0.8480	770/3082
Scheduled Sparse 25%	0.8920	770/3082

Scheduled sparse ANP is substantially more stable than random sparse ANP at H1024: the random policy drops 5.89 pp from the full-ANP baseline, while the scheduled policy drops only 1.49 pp. This difference is notably larger than in DANP, where random and scheduled were nearly interchangeable (-0.84 pp vs -0.76 pp). Without the decorrelation update reducing inter-node dependencies, random selection is more likely to miss important nodes for multiple consecutive steps, resulting in higher-variance gradient estimates. The round-robin schedule prevents this by guaranteeing every node is visited within H/k steps. From an implementation standpoint, the scheduled policy also requires no comparator logic: the active window advances by a fixed stride k , implementable as a single modular counter. These results indicate that the scheduled policy is the more appropriate choice for sparse ANP.

3.3 CIFAR-10 ANP

Under the current hyperparameter configuration, both full ANP (0.1043) and Scheduled Sparse ANP 25% (0.1049) remained near chance level on CIFAR-10. Because the full baseline itself did not converge, the sparse result cannot be interpreted independently. No conclusion regarding sparse ANP on CIFAR-10 is drawn from these experiments.

4 Gradient Distribution Comparison

To investigate why adaptive node-selection policies did not consistently outperform random or scheduled sparse selection, the concentration of the node perturbation effective-error signal was measured for both ANP and DANP. Per node i , the effective-error magnitude is:

$$g_i \propto \frac{\Delta a_i \cdot \Delta L}{\|\Delta \mathbf{a}\|^2} \quad (1)$$

Three concentration statistics were recorded over 5 epochs (3 seeds, [1024,1024,1024], full algorithm, MNIST):

Table 5: Gradient distribution statistics at epoch 5 (mean across 3 seeds, MNIST [1024,1024,1024]). CV: coefficient of variation of $|g_i|$; Top 10%: fraction of total $|g|$ energy in the top-10% of nodes; Gini: Gini coefficient of $|g_i|$.

Algorithm	CV	Top 10% Energy	Gini
ANP	0.296	0.149	0.165
DANP	0.281	0.144	0.156

ANP shows marginally higher concentration than DANP across all three metrics (CV: +0.015, Top 10%: +0.005, Gini: +0.009), consistent with the decorrelation update making the signal slightly more uniform. However, both algorithms remain in the same diffuse regime: CV is well below 1, and the top-10% of nodes carry only about 15% of the total signal energy, versus

50% or more that would indicate strong concentration. These results are consistent with the observation that adaptive node-selection policies are unlikely to provide a consistent accuracy advantage over random or scheduled sparse selection for either algorithm.

5 Figures

Figures 1–4 plot DANP data from Table 2. ANP data are summarised in Tables 3 and 4.

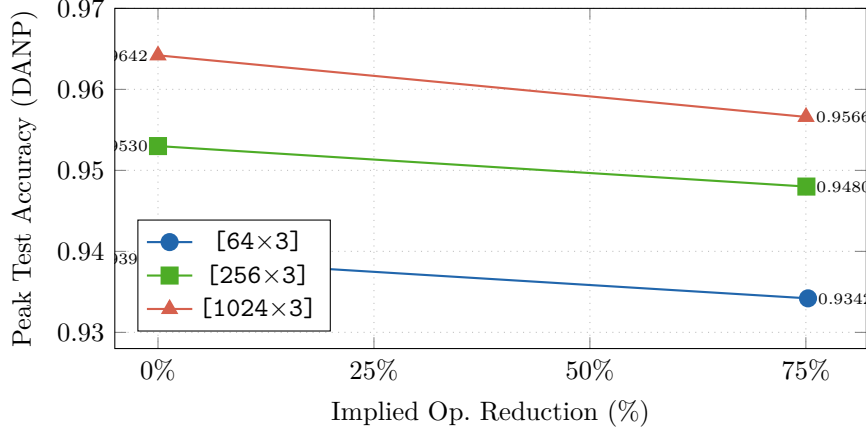


Figure 1: DANP: accuracy vs. implied perturbation operation reduction. Each line connects full DANP (0% reduction) to Scheduled 25% (≈75% operation reduction) for one architecture.

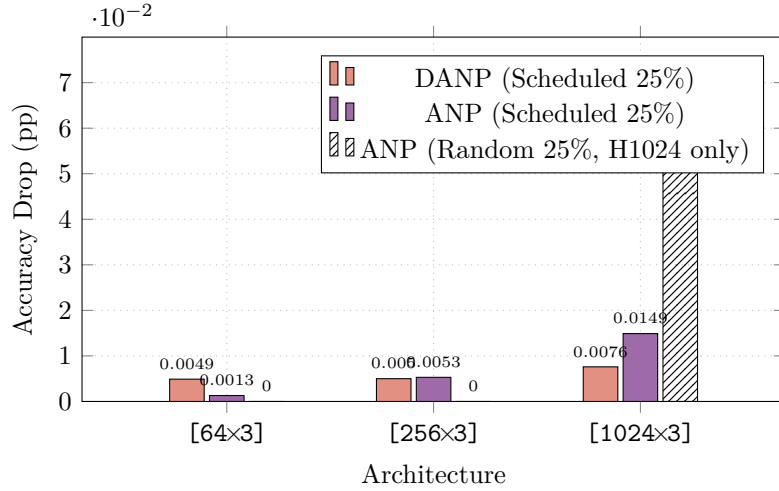


Figure 2: Accuracy drop at ≈25% active nodes: DANP vs. ANP. For ANP at H1024, the random and scheduled policies are shown separately to illustrate the policy sensitivity discussed in Section 3.

6 Conclusions

- Sparse Scheduled ANP at H64 and H256 preserves nearly all of the full-ANP accuracy with only 25% active nodes (drops of 0.13 pp and 0.53 pp respectively), suggesting that sparse ANP without decorrelation is a viable approach at smaller widths and may be worth pursuing in hardware designs where decorrelation is not yet implemented.

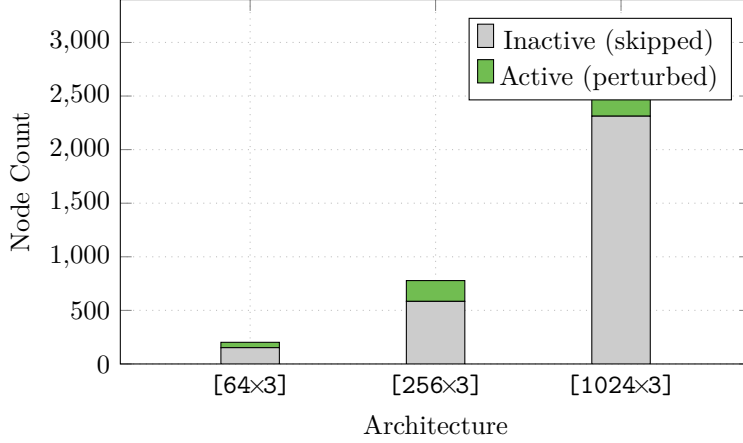


Figure 3: Active (perturbed) vs. inactive (skipped) nodes per architecture under Scheduled 25% (same mask for ANP and DANP).

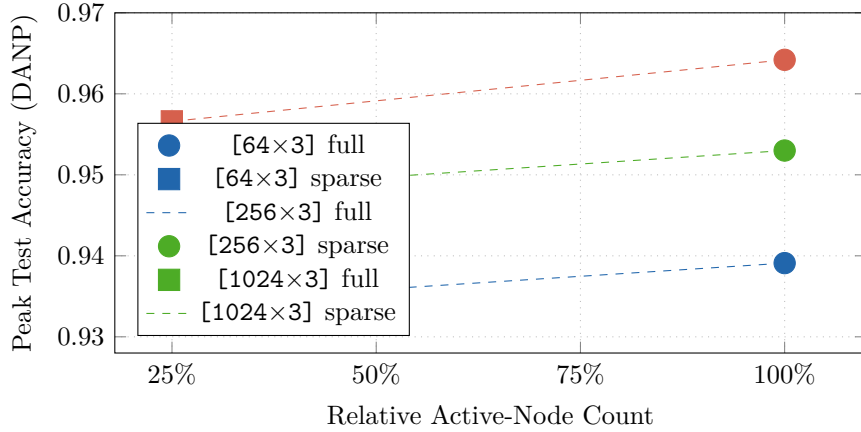


Figure 4: DANP: relative active-node count vs. test accuracy (circles: full; squares: Scheduled 25%). Dashed lines connect full and sparse runs of the same architecture.

- Scheduled sparse selection is substantially more stable than random sparse selection for ANP at H1024 (-1.49 pp vs. -5.89 pp), a difference not observed in DANP. The round-robin schedule also requires no sorting or ranking logic, making it the more practical choice for sparse ANP.
- DANP remains stronger than ANP at H1024 under sparse conditions (-0.76 pp vs. -1.49 pp), consistent with the decorrelation update providing a learning-efficiency advantage at larger scale.
- Gradient-distribution measurements indicate that the effective-error signal is in a diffuse regime for both ANP ($CV = 0.296$) and DANP ($CV = 0.281$). These results suggest that adaptive node-ranking strategies are unlikely to provide consistent gains over simple random or scheduled selection for either algorithm.
- CIFAR-10 ANP results are not interpretable under the current hyperparameter configuration, as the full baseline did not converge.